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RAG



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What is RAG in AI World?

Memory plays a key role in our lives at multiple levels, including routine activities, conversing with others, socialization, learning, professional work, decision-making, and communication, to name a few.

Memory is the vital element in human survival.

Without it, the world wouldn't be what it is today, with so many advancements improving human life.

We — the humans — built machines, airplanes, rockets, and many more.

First, we built machines mimicking human activities. Now our lives are mostly dominated by gadgets and machines. The technology is so advanced that we wouldn't have thought about it.

Now, we live in an AI world. The primary buzzword in the AI world is RAG.

What is RAG in this AI world?

RAG stands for **Retrieval-Augmented Generation**. It is a technique used to improve AI outputs by grounding them in specific, proper, reliable data.

RAG is the *process of generating clean, and reliable data*.

The RAG Process:

- **Retrieve:** Find useful, relevant information from external sources (like documents or databases).
- **Augment:** Add that retrieved information to the AI's existing knowledge base for a specific query.
- **Generate:** Create a more accurate, context-aware response based on the combined information.

How does Retrieval-Augmented Generation work?

RAGs operate with a few main steps to help enhance generative AI outputs.

- **Retrieval and pre-processing:** RAGs leverage powerful search algorithms to query external data, such as **web pages, knowledge bases, and databases**.

Once retrieved, the relevant information undergoes pre-processing, including tokenization, stemming, and removal of stop words.

- **Grounded generation:** The pre-processed retrieved information is then seamlessly incorporated into the *pre-trained LLM*.

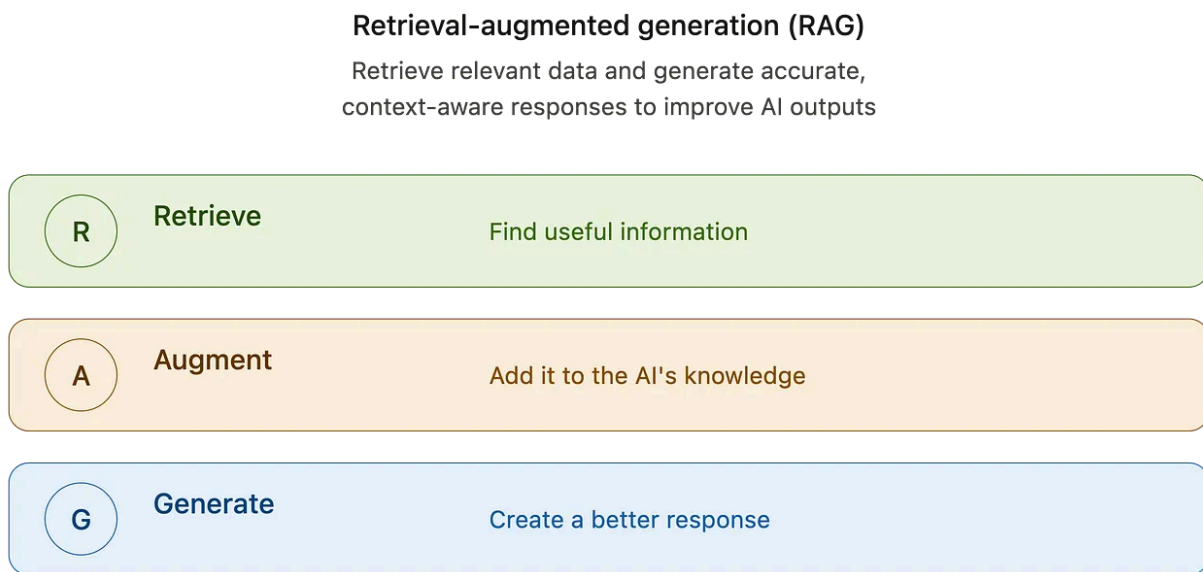
This integration enhances the *LLM's context*, providing it with a more comprehensive understanding of the topic. This augmented context

enables the LLM to generate more *precise, informative, and engaging* responses.

Why use RAG?

RAG offers several advantages augmenting traditional methods of text generation, especially when dealing with factual information or data-driven responses.

Once retrieved, the relevant information undergoes pre-processing, including tokenization, stemming, and removal of stop words.



Here are some key reasons why using RAG can be beneficial:

Access to fresh information

LLMs are limited to their pre-trained data. This leads to outdated and potentially inaccurate responses.

RAG overcomes this by providing up-to-date information to LLMs.

Factual grounding

LLMs are powerful tools for generating *creative* and *engaging text*, but they can sometimes struggle with factual accuracy. This is because LLMs are trained on massive amounts of text data, which may contain inaccuracies or biases.

Providing “facts” to the LLM as part of the input prompt can mitigate “*gen AI hallucinations*.”

The crux of this approach is ensuring that the most relevant facts are provided to the LLM, and that the LLM output is entirely grounded on those facts.

And also answering the *user’s question* and adhering to *system instructions* and *safety constraints*.

Using Gemini’s **long context window (LCW)** is a better way to provide source materials to the LLM.

If we need to provide more information than fits in the LCW, or to **scale up performance**, we can use a RAG approach to reduce the number of tokens, saving you **time** and **cost**.

Vector databases store documents as embeddings in a high-dimensional space. This feature allows — fast and accurate retrieval based on semantic similarity.

Multi-modal embeddings can be used for *images*, *audio* and *video*, and more — and these media embeddings can be retrieved alongside **text embeddings** or **multi-language embeddings**.

Advanced search engines like **Agent Search** on Gemini Enterprise Agent Platform use **semantic search** and keyword search together (called *hybrid*

search), and a **re-ranker** which scores search results — to ensure the top returned results are the most relevant.

Additionally searches perform better with a **clear, focused query** without misspellings; *so prior to lookup, sophisticated search engines will transform a query and fix spelling mistakes.*

Relevance, accuracy, and quality

The retrieval mechanism in RAG is critically important.

We need the best semantic search on top of a curated knowledge base to ensure that the retrieved information is relevant to the input query or context.

If the retrieved information is irrelevant, the final generation could be grounded but off-topic or incorrect.

By **fine-tuning** or **prompt-engineering** the LLM to generate text entirely based on the **retrieved knowledge**, RAG helps to minimize **contradictions** and **inconsistencies** in the generated text.

This significantly improves *the quality of the generated text*, and *improves the user experience*.

The *Model evaluation in Gemini Enterprise Agent Platform* by Google, now scores LLM generated text and retrieved chunks on metrics like:

- “**coherence**,” “**fluency**,” “**groundedness**,”
- “**safety**,” “**instruction_following**,”
- “**question_answering_quality**,” and many more.

These metrics are important and help you measure the grounded text you get from the **LLMs**.

Implementing the above evaluations creates a baseline measurement and can be optimized for RAG quality by **configuring the search engine, curating the source data, improving source layout parsing or chunking strategies, or refining your question prior to search.**

A **RAG Ops**, metrics driven approach like this will help us doing hill climb to *high quality RAG* and **grounded generation**.

One thing I can tell you: there is so much of information and usage of new keywords — along with the real meaning — is the key factor for approaching any context data.

Finally:

Your data and your use cases are what differentiate and what you are building with **Gen AI**.

RAG and grounding bring your data to LLMs efficiently and scalably.

It is difficult to understand or process the data to gain knowledge, if anyone is not able understand the context, keywords, and their appropriate meanings — along with the usage.

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